

Problem Set 13
Due: May. 31, 2021

1. Section 5.4 Problem 9.

During each day, the probability that your computer's operating system crashes at least once is 5%, independent of every other day. You are interested in the 45 crash-free days out of the next 50 days.

- (a) Find the probability of interest by using the normal approximation to the binomial.
- (b) Repeat part (a), this time using the Poisson approximation to the binomial.

2. Section 5.4 Problem 10.

A factory produces X_n gadgets on day n , where the X_n are independent and identically distributed random variables, with mean 5 and variance 9.

- (a) Find an approximation to the probability that the total number of gadgets produced in 100 days is less than 440.
- (b) Find (approximately) the largest value of n such that

$$\mathbf{P}(X_1 + \cdots + X_n \geq 200 + 5n) \leq 0.05$$

- (c) Let N be the first day on which the total number of gadgets produced exceeds 1000. Calculate an approximation to the probability that $N \geq 220$.

3. Section 5.4 Problem 11.

Let $X_1, Y_1, X_2, Y_2, \dots$ be independent random variables, uniformly distributed in the unit interval $[0, 1]$, and let

$$W = \frac{(X_1 + \cdots + X_{16}) - (Y_1 + \cdots + Y_{16})}{16}$$

Find a numerical approximation to the quantity

$$\mathbf{P}(|W - \mathbf{E}[W]| < 0.001)$$

4. It is well known that infants born to mothers who smoke tend to be small and prone to a range of ailments. It is conjectured that also they look abnormal. Nurses were shown selections of photographs of babies, one half of whom had smokers as mothers; the nurses were asked to judge from a baby's appearance whether or not the mother smoked. In 1500 trials the correct answer was given 910 times. Is the conjecture plausible? If so, why?